Biospectra : Vol.9(1), March, 2014. pp 129-134
An International Biannual Refereed Journal of Life Sciences

Interdisciplinary Articles

ISSN : 0973-7057

Rearing performance of eri silk worm *(Philosamia ricini)* on Girija caster variety grown in Krishna river basin of Sangli district.

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Received , 20th January, 2013 ;Revised: 05th February, 2014

Abstract : Caster *(Ricinus communis* L*)* is the primary food plant of eri silk worm *Philosamia ricini* Hutt.is cultivated in Maharashtra in about 25,000 hectares .mainely for its non-edible oil seed production purpose . In sangli district sugarcane, soybean, turmeric and grapes are the main cash crops, however caster is cultivated on bandhas and especially in drought affected tahsils viz .Atpadi ,Jat, Khanapur, Vita, Miraj (east), Tasgaon and Khanapur. .Ericulture will be a boon for small farmers to enhance their income from caster. In the present study it was observed that Girija caster variety is prevalent in sangli district. The rearing batches fed on Girija caster variety and the results of experiment was carried out on growth, development ,economic cocoon parameters and grainage parameters of eri silk worm *(Philosamia ricini).* It was observed that the weight of mature larva was 4.859gm and total instar duration was 16.320 days, the larval period was 21.20 days and the pupal period was about 18 to 21 days the survivability of *Philosamia ricini* was 97.42%, whereas the rate of rearing ,cocoon characteristics and grainage parameters were significantly maximum. The study indicates that there is scope for further expansion of ericulture to improve socioeconomic conditions of especially drought affected farmers.

Key Words: Eri, castor ,Girija, Ericulture.

INTRODUCTION

India is a known to vast variety of silk secreting fauna which includes an amazing diversity of silk moths. This has enabled India to achieve the unique distinction of being a producer of all the five commercially traded verities of natural silk namely, mulberry, tropical tasar, oak, eri and muga silk. India stands second in world with respect to eri silk production next only to china by producing 1485 metric tonnes. In India eri sector was confined to the north East and congruent eastern India till 3 decades ago. Now it is a vocation for rural reconstruction in non traditional areas too since it involves low investment and high output source of employment and income. Presently it is practiced in 17 indian states contributing to 11% of the total raw silk production in India (Monica chaudhari 2009)

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Rajaram and Samson (1991) studied about the rearing performance of eri silk worm. The study confirm that castor is the most suitable plant. Jadhav A.D. (2012) observed that the possibility of introduction of ericulture lies in the utilization of leafy rich high yielder castor varieties like local Green, local red, GCH- 4 by Judicious conversion of the leaves into silk and beans oil. The castor crop (*Ricinus communis*) is well known as oilseed crop but its potentiality for rearing eri silkworms is not tapped properly in this region. It has further significance being very hardy, resistant to drought and insect pest and very much suited as rain fed crop while many crops fail due to erratic rainfall/ intermittent dry spells. It is also substantiated through research findings that ericulture do not lower the yield of oilseed of castor and its cultivation is cost effective than mulberry sericulture. If due weightage is given to this crop for ericulture, the economy of so called marginal and poor farmers associated with marginal lands will definitely increase. Not only farmers

will be benefited but the availability of raw material (eri cocoons and silk wastes) for agro-based cottage industries will be ensured form neighboring villages at relatively cheaper rate.

Mulbery and tasar silk production is well known to the state, but extensive trials with respect to ericulture is yet to be demonstrated. Sericulture did not gain importance in Maharashtra due to alternative cash crops like sugarcane, cotton, grapes, onion and mango and in spite of all these Maharashtra emerging as a contender for sericulture state among the non-traditional states of India.

The agro-ecology of Maharashtra is well suited to caster on a large scale. Castor grows in almost any soil with the exception of very heavy clay soils and those with poor drains. Castor is one of the mixed crop espessially taken on bandhas in Atpadi, Khanapur, Vita, Walwa, Jat, Kavthemanhkal tahasils of Sangli district.

MATERIALAND METHODS

Eri seeds were brought from Central silk Board ,Assam.A light chief house was constructed in premises of K.N.P.college walwa.The plastic rearing trays and racks were employed and regularly disinfected with 2 percent formalin. Both caster plant insect pest and pests of eri silkworm were prevented. During the study period castor crop was raised and maintained .Few castor varieties like Aruna,Girija,V.I.9 were selected for plantation and rearing trials The eggs were incubated at 25+-1°c For hatching. Later eri larvae were mass reared up to second moult by using plastics tray of the size 3'X2'X4" by recommended pacing of rearing .For rearing of the late age worms shelves prepared out of bamboo and local materials were used .Foliage form planted castor varieties was used for feeding the silkworms. All agronomic practices were followed as per package, Anon (1994).The data on parameter such as growth ,development, economic cocoons parameter and grainage parameter of eri silk worm were studied . An entire leaves were fed to 3rd and 4th instar larvae. First and second instar stages were fed with chopped caster leaves .A high resolving camera was operated for the photographic work.

RESULTS

Eri silk worm rearing was practiced on Girija variety of castor during the period of November-December and January-February.

The results of experiment carried out on growth, development and economic cocoon parameters of Eri Silk worm *Philosamia ricini* are presented here.

Table – I. Performance of different instars of Eri silk worm on Girija castor plant

Instar of eri-silk worm	I	II	III	IV	V
Larval weight in (g)	0.020	0.482	1.705	2.669	6.729
Larval duration (days)	1.883	2.632	2.449	3.512	5.603

Table no. I indicate weight and duration of I to V instar of eri-silk worm. However the weight of mature larva was 4.859(g) and the total instar duration was recorded during rearing period November-December 16.320 days.

The larval period recorded during November-December was 21.20 days, the pupal period extends for about 18 to 21 days. It was found that eri silk worm completes its life cycle in 47.20 days. The incubation period was about 8 days .

It was observed in the above experiment the survivability of *Philosamia ricini* on Girija variety of castor was superior 97.42% . The effective rate of rearing was significantly maximum (93.82%).



Fig. 1. First instar larvae of eri feeding on caster leaves



Fig. 2. Second instar larvae on caster leaves



Fig.3. Third and fourth instars of eri silk-worm feeding on caster leaves.



Fig.4. Late age third and fourth instars of eri silk-worm.



Fig.5. Fourth instar of eri on caster feed,.



Fig.6. Dialated fifth instar voraciously feeding on caster leaves. Note alternate blue and white bands on mature silk-worm.

Cocoon characteristics

Cocoon characteristics of Eri Silk Worm (*Pilosomia Ricini)*On Girija on Caster Variety.

Table no. II indicates cocoon weight, pupal weight, shell weight and shell ratio of eri silk worm *philosamia ricini*. It was observed that all the cocoon characters were significantly higher.

Cocoon weight	Pupal weight	Shell Weight	shell ratio
2.38 (g)	2.098(g)	0.282 (g)	11.98%

Grainage parameters

The grainage parameters moth emergence was recorded 82.86%, the average fecundity was recorded and is highest 298.00. The hatching percentage was also found highest 92.1%.

The results of present investigation on growth, development and cocoon parameters of eri, silk worm *Philosamia ricini* indicates that the introduction of ericulture in the district will be opportunity to get additional income to castor growers. The nutritional quality of caster leaves influence the growth and development of eri silk worms, cocoon yield & quality. Eri silk worm is said to be hardy and less susceptible for diseases. The fact has been confirmed in present rearing trials by getting good yield.

The observations of present investigation indicates that fresh quality tender leaves supplied to silk worms increase the larval weight and other commercial parameters. It has been reported that caster seed yield was not affected although 20 to 25%of the total leaf yield can be harvested. Similar kind of observation was reported by Isa et al.(1994) and Sannappa et al .(2002) that caster plant can be defoliated up to 25 per cent and in turn these leaves can be utilized for eri silkworm rearing. It has been also reported that 25% of the total leaf yield can be harvested and used for eri cocoon production without affecting the castor seed yield (Devaiah et al, 1984 and Siddiqui et al, 1983). Similarly, the research findings of Directorate of oil seeds research, Hyderabad revealed that defoliation to an extent of 25-30 percent does not affect the yield of castor as it has tremendous regenerative capacity (Teotia et.al,2003).According to Suryanarayana et al.(2003),Gujarat, Andhra Pradesh, Rajastan, Tamil Nadu,Orissa, Karnataka and Maharashtra states have been cultivating caster on large scale for production of seed to extract oil.These states can successfully utilize 30-40 per cent of caster leaves for obtaining additional income through eri silkworm rearing without impairing caster seed yield. (D. Chandrappa et al, (2012).The ideal range of temperature for the growth of eri silk worms is from 20^{0c} to 40^{0c} however, increase in temperature beyond 35^{0c}causes less spinning, mortality of larvae and pupa and poor moth emergence and sterility at adult stage (Sahu, et. al, 2006).

The environmental conditions such as temperature and humidity are best suited for ericulture in Sangli district. The initial rearing trials of eri silk worms in Sangli district are encouraging. The castor crop is very hardy , resistant to draught and insect pest and very much suited as rain fed crop while many crops fail due to erratic rainfall intermittent dry spells, which was found in especially Atpadi, Vita, khanapur, Miraj, Tasgaon,Kawatemahankal and kadegaon tahasils of sangli district.

There is need to encourage raising systematic plantation of eri food plants. The promoting activities from directorate of sericulture, Maharashtra for caster farming and development of ericulture are essential. It will be an dual income to castor growers in the drought prone tehsils of Sangli district by utilizing only the family members of farmers.

Potential, Strength and challenges of ericulture industry in India R and D achievement like development of indigenous castor varieties with highest leaf yield and the new bivolatine eri silkworm hybrids eminently suited to

the tropical regions of the country. Farmers friendly technologies,cost effective new packages of practices for cultivation of plants, rearing and realing coupled with huge natural and man-made resources and trained man power clearly indicates the future prospects of sericulture industry ,to emerge as a promising indicator of economic development for the upliftment of the socially deprived communities and the down trodden.



Fig.7. Fifth instar larvae of eri silk-worm feeding on casrer leaves



Fig.8. Note the change in colour of different stages of silk worm.

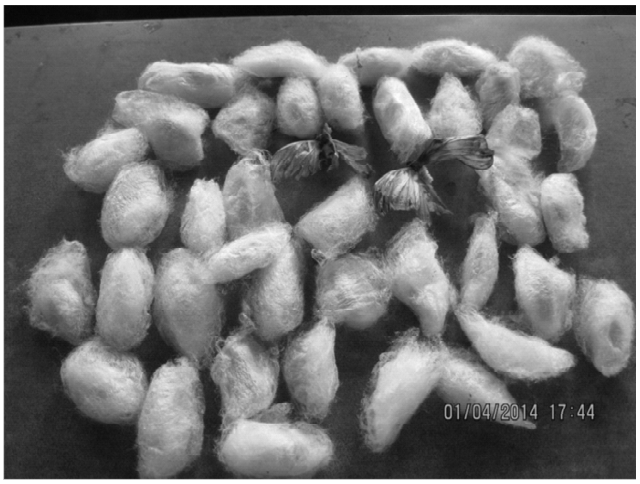


Fig.9. Eri silk moth enmergance from cocoon

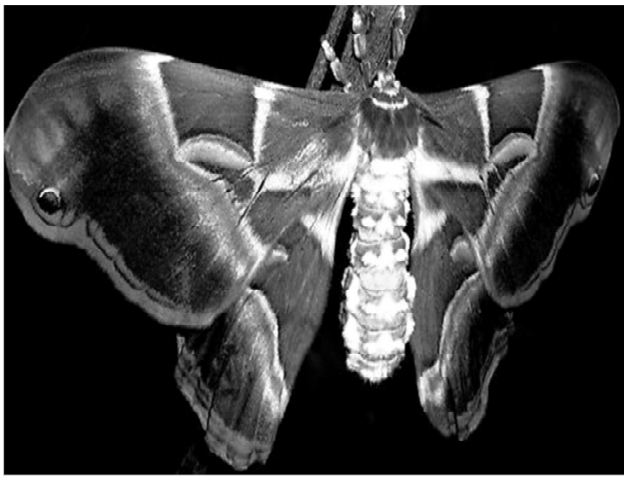


Fig.10. Adult eri silk moth.

ACKNOWLEDGEMENT

Author is thankful to UGC New Delhi for providing financial assistance

REFERENCES

1. **D.Chandrappa, Sannappa Ramkrishna Naika, B.,K., Manjunath, G. and Govindan, R. 2012.** Differential performance of Castor genotypes in seed yield and eri cocoon production and their economic analysis under rained condition. Global journal of biology, agriculture and health science Vol.1(2):1-3.

2. **Devaiah, M.C., Rajashekharagouda, R., Suhas,Y.And Govindan, R.,1985**. Growth and silk production in Samia Cynthia ricini Boisduval

3. **Monica Chaudhari 2009**. Ericulture:A Bioprospecting for supplimeting livelihood, enviorment and food security in rural and semi -urban India .Eri research cell Dec 30,2009.

4. **Rajaram and samson,Nm.V., 1991.** Cinnamomum glanduliferum messn as a secondary host plant for eri silkworm Samia cynthia ricini Hutt. Indian J.seric.,30(1):64-65.

5. **Jadhav , A. D., Hajara, T. N. Prasad Jagdish and Rode,S.A. 2004**. Prospects of Ericulture in Maharashtra .Indian silk,(9):15-16.

6. J**adhav ,A. D, 2012.** Rearing performance of eri silkworm *Philosamia ricini*(Hut.)on different varieties of caster grown under Vidarbha region, Biospectra.7(1),pp.45-50.

7. **Isa, M.D.,Yadav, G.S., Prasad, D.N. and Sinha, S.S. 1994**.Effect of leaf harvest on the seed yield of castorRicinus communis L. Indian J. Seric.33:76-77.

8. **Sannapp, B., Jayaramaiah, M., Govindan, R. and Chinnaswamy, K.P .2002**.Advances in Ericulture. Seri scientific pub.,Banglore,P.144.

9. **Sahu, M., Bhuyan, N., & Das, P.K. 2006.** Eri Silworm,Samia Ricini(Lepiddotera: Saturnidae) Donovan,Seed production during summer in Assam .In :Proceeding of Regional seminar on "Prospects and problems of sericulture as an economic enterprises in North West India 2006" ,Dehradune, India,(490-493).

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